



University of Piraeus
School of Information and Communication Technologies
Department of Digital Systems

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Wireless and Mobile Network Security

Case 4

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Abstract

Examination of the security of IEEE 802.11 wireless networks through analysis in a Kali Linux environment. Analysis of management frames (Beacon, Probe Request/Response), study of the WPA2 4-way handshake mechanism, and practical implementation of an Evil Twin attack.

Abstract

Security assessment of IEEE 802.11 wireless networks through analysis in a Kali Linux environment. The study performs a detailed analysis of Beacon and Probe frame structures, as well as the 4-way handshake process using Wireshark. The second part involves the implementation of an Evil Twin attack and the creation of a Rogue Access Point, demonstrating the inherent vulnerabilities of the 802.11 protocol against Layer 2 attacks.

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Introduction

The security of wireless communications is one of the most critical areas of modern computing. This assignment examines the inner workings of the WPA2 protocol through frame analysis and implements an Evil Twin attack.

The goal of the assignment is practical familiarity with the methodologies for analysing and attacking wireless networks.

First, an analysis of management frames is carried out — the frames used for the control and management of wireless network infrastructure.

- Beacons: Packets periodically broadcast by the Access Point (AP) — the central device acting as a bridge between the wireless and wired network. They contain the network name (SSID), the supported data rates, and the security parameters.
- Probes: Detection messages exchanged between devices and APs during the search for available networks.

In parallel, the key exchange process known as the 4-way handshake is studied — a mutual authentication and encryption key establishment mechanism between the client and the access point.

For personal data protection purposes and following security best practices when publishing this report, sensitive identifying details have been intentionally obscured in the screenshots.

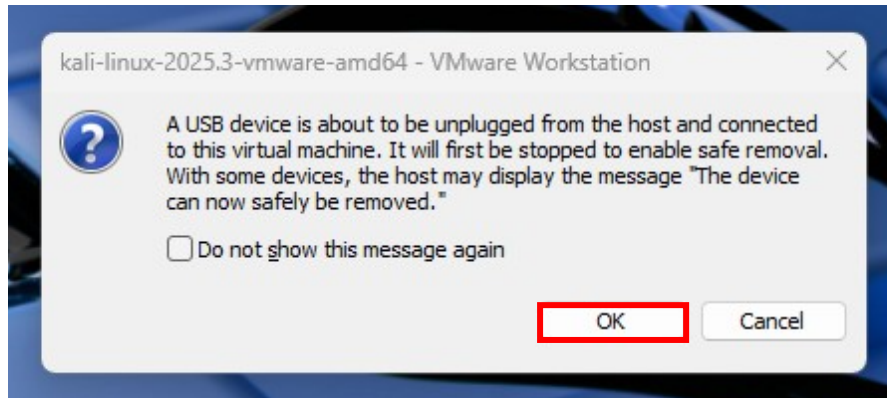
Practical part of the assignment



Regarding the first step of connecting the equipment to VMware, the process begins by inserting the TP-Link TL-WN722N adapter and connecting it exclusively to Kali Linux.

When the device is connected, VMware displays a standard informational message confirming that the TP-Link card has been released from the host machine.

The



iwconfig command was executed to verify successful recognition of the wireless interface.

Kali Linux successfully recognised the adapter as wlan0 in Managed mode. The airmon-ng check kill command was executed to terminate interfering background processes.

```
(kali@kali)-[~]
$ sudo airmon-ng check kill
[sudo] password for kali:
Killing these processes:

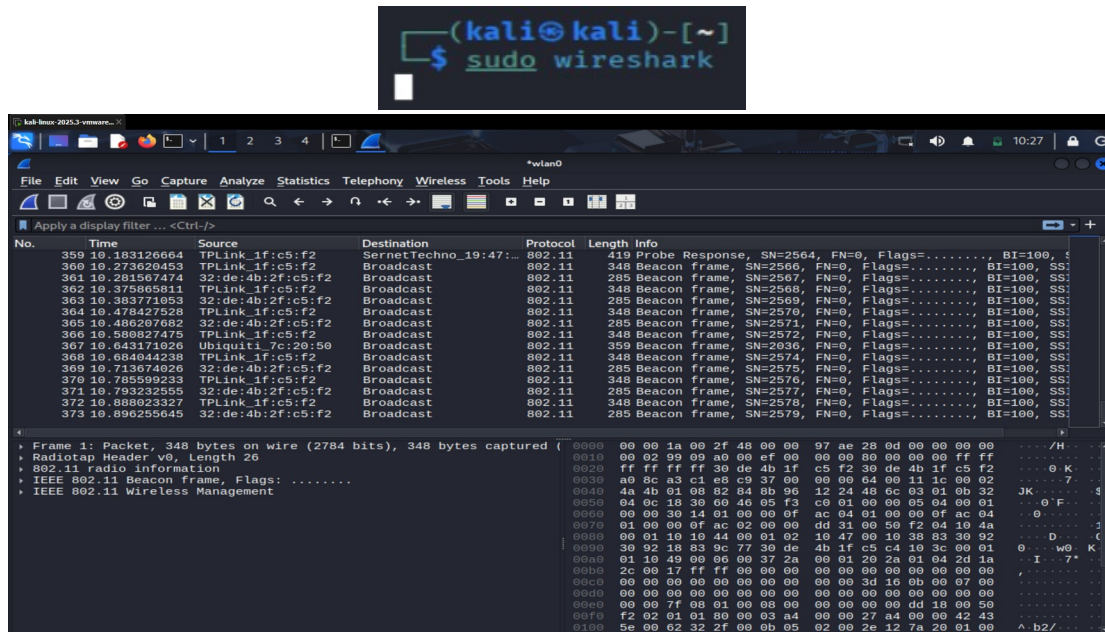
    PID Name
    2347 wpa_supplicant

(kali@kali)-[~]
$
```

The system terminated the wpa_supplicant process and the adapter was switched to Monitor Mode.

Question A

To analyse the negotiation messages, wireless traffic was captured using airodump-ng.

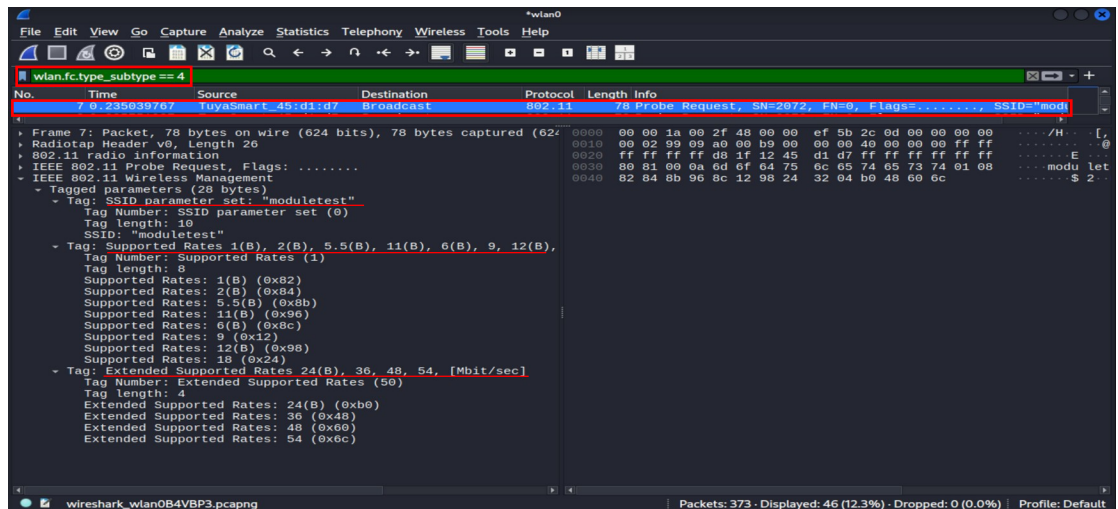


In the first stage of analysis, the filter `wlan.fc.type_subtype == 8` was applied to isolate Beacon frames.

- **Network Name (SSID):** The SSID parameter set field confirms the wireless network name is 'JK'.
- **Operating Channel:** The DS Parameter set field indicates the AP is broadcasting on channel 11.
- **Security Parameters (RSN Information):** The RSN Information section confirms the network uses modern security protocols.
- **Auth Key Management (AKM) List:** Set to PSK (Pre-Shared Key), indicating that authentication is performed via a pre-shared passphrase — the standard home network method.
- **Pairwise Cipher Suite:** Set to AES (CCM), meaning the CCMP algorithm is used for encrypting unicast traffic.
- **Group Cipher Suite:** Also set to AES (CCM), determining that multicast and broadcast traffic is encrypted with the same algorithm.

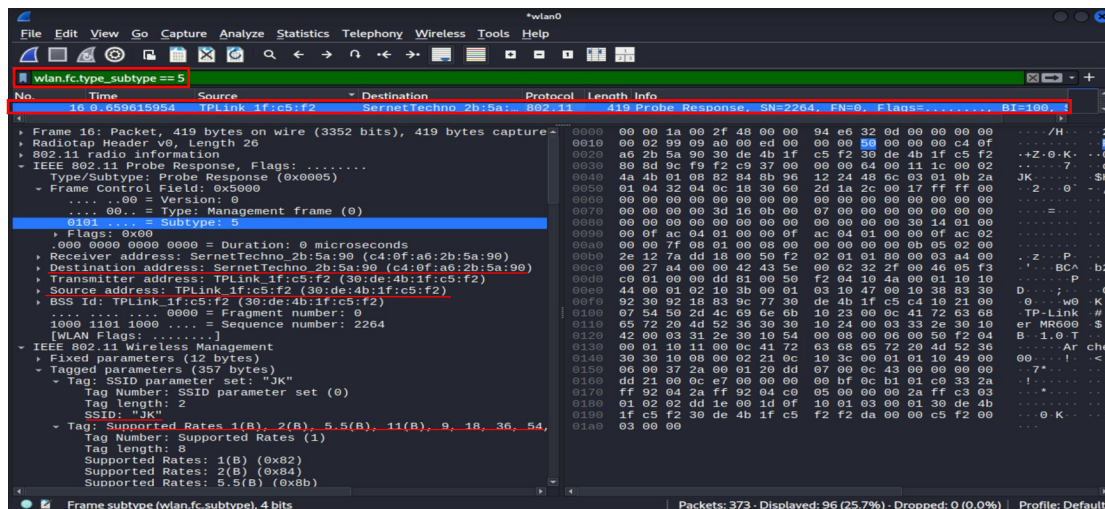
Next, a Probe Request message was examined, isolated using the filter `wlan.fc.type_subtype == 4`.

- **Targeted Network Search (SSID):** The SSID parameter set field shows the device is actively searching for the 'JK' network.
- **Supported Rates:** In the 'Supported Rates' and 'Extended Supported Rates' fields, the device advertises its supported data rates.



Finally, a Probe Response message was recorded, isolated with the filter `wlan.fc.type_subtype == 5`.

- **Communication Addresses:** The response originates from the Access Point (Source) and is directed towards the specific client device.
- **Network Confirmation and Connection Parameters:** The SSID parameter set field confirms the availability of the 'JK' network.



Questions B and C

To record the secure connection establishment process (4-way handshake), the `eapol` filter was applied in Wireshark.

1. PTKSA Establishment (Messages 1 and 2):

The initial exchange concerns the creation of the Pairwise Transient Key (PTK), which protects unicast traffic.

- **Message 1:** In the first stage of the negotiation, the AP sends the ANonce — a random number (nonce) that the client will use to generate the PTK.
- **Message 2:** The client responds with its own SNonce and a Message Integrity Code (MIC), proving it possesses the correct pre-shared key.
 - **WPA Key Nonce:** The recorded hexadecimal value is the SNonce (Supplicant Nonce) — the client's random parameter.
 - **WPA Key MIC:** Unlike the first message, this packet's field contains a valid integrity check code — proof that the client knows the shared secret.

With the successful receipt of this message, both parties now have all the necessary elements to independently calculate the same PTK.

2. GTKSA Establishment (Messages 3 and 4)

Once authentication completes successfully, the process moves to protecting broadcast/multicast traffic via the Group Transient Key (GTK).

- **Message 3:** The AP sends the client the GTK, encrypted with the PTK established in the previous exchange.
- **Message 4:** The client sends a final confirmation message (Unicast, MIC) to the AP, confirming successful receipt and installation of the GTK.

Analysis and implementation of Evil Twin attacks: Creation of a Rogue Access Point and Captive Portal

In the final part of the laboratory exercise, an Evil Twin attack was attempted. The attack involves creating a fake AP identical to the legitimate one, sending Deauthentication frames to force the client to reconnect to the fake AP, and capturing the credentials entered via a Captive Portal.

Initially, the automated framework airgeddon was used. Through its built-in interface, it provided guided access to network scanning functions, Monitor Mode activation, and handshake capture.

```

kali@kali: ~
Session Actions Edit View Help
***** Select target *****

```

N.	BSSID	CHANNEL	PWR	ENC	ESSID
1)		6	25%	WPA2	To A34 του/της Bayy
2)		1	25%	WPA2	VODAFONE_WIFI_108-15627
3)		6	26%	WPA2	(Hidden Network)
4)		7	27%	WPA2	COSMOTE-296767
5)		11	28%	WPA2	(Hidden Network)
6)		13	29%	WPA2	COSMOTE-734612
7)		12	29%	WPA2	COSMOTE-757622
8)		9	29%	WPA2	COSMOTE-bhdb25
9)		11	29%	WPA2	Lakis
10)		6	30%	WPA2	Dermentzopoulos
11)*		1	34%	WPA2	VODAFONE_H268Q-9410
12)		6	37%	WPA2	COSMOTE_3J5285_2.4G
13)		1	38%	WPA2	Huawei_Xfv4NM
14)		8	40%	WPA2	VODAFONE_H268Q-3372
15)*		4	56%	WPA2	VODAFONE_H268Q-4971
16)		11	67%	WPA2	(Hidden Network)
17)	30:DE:4B:	11	67%	WPA2	JK

```

(*) Network with clients
Select target network:
>

```

In order to enable subsequent password verification at the Captive Portal, the 4-way handshake had to be captured first.

During the attempt to automatically deploy the Rogue AP and Captive Portal, technical limitations were encountered: the hostapd tool failed to create a virtual AP on the available hardware.

To bypass the hostapd limitations and demonstrate the feasibility of the attack, the process was repeated manually using a separate physical access point configured as a Rogue AP.

```

(kali@kali)-[~]
$ sudo airmon-ng check kill
Killing these processes:
    PID Name
    67826 wpa_supplicant

(kali@kali)-[~]
$ sudo airmon-ng start wlan0

```

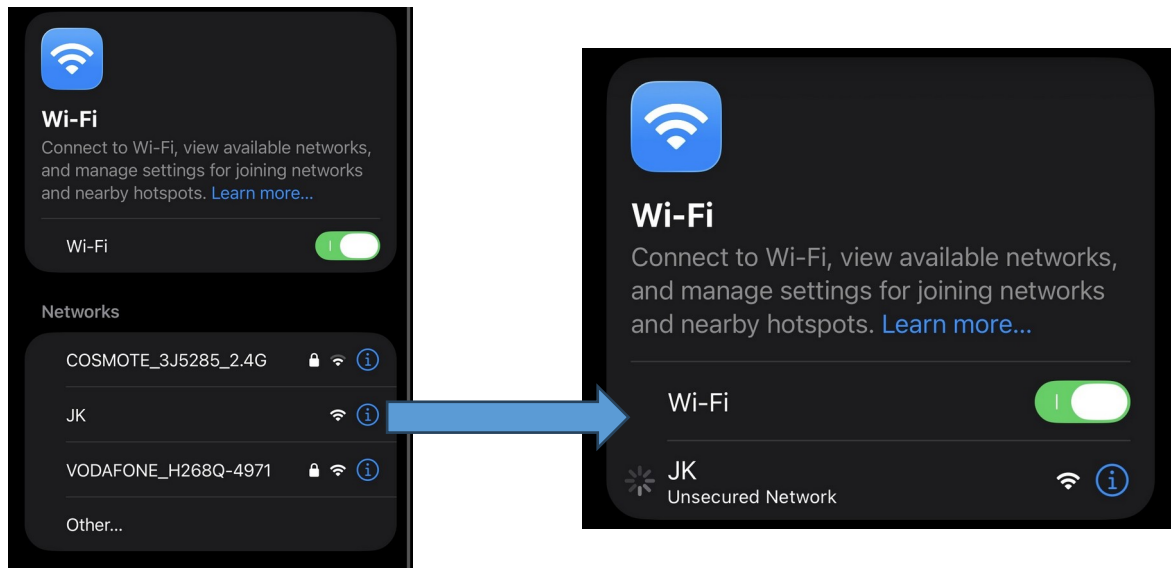
PHY	Interface	Driver	Chipset
phy1	mon1	rtl8xxxu	TP-Link TL-WN722N v2/v3 [Realtek RTL8188EUS]
phy1	wlan0	rtl8xxxu	TP-Link TL-WN722N v2/v3 [Realtek RTL8188EUS]

```

(mac80211 monitor mode already enabled for [phy1]wlan0 on [phy1]10)

(kali@kali)-[~]
$ sudo airbase-ng -e "JK" -c 11 wlan0
ioctl(SIOCSIWMODE) failed: Device or resource busy
15:25:23 Created tap interface at0
15:25:23 Trying to set MTU on at0 to 1500
15:25:23 Trying to set MTU on wlan0 to 1800
15:25:23 Access Point with BSSID 30:DE:4B: started.
15:27:58 Client BE:FC:A4: associated (unencrypted) to ESSID: "JK"

```



Conclusion

This exercise confirmed in practice that the security of a wireless network does not depend exclusively on the strength of the WPA2 passphrase.

However, the implementation of the Evil Twin attack revealed the protocol's 'Achilles heel': the inability to authenticate the Access Point itself. Nothing in the standard prevents an attacker from creating an identical access point.

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